The Quadratic Formula

Name _____

Algebra 2 Lesson 2-6 (fast track) Date _____ Period _____

Learning goal: I can solve quadratic equations using the Quadratic Formula and use the discriminant to predict the number and type of solutions.

Language goal: I can explain how many solutions a quadratic equation has and what type they are, using the words discriminant, radicand, real, and non-real.

On the quiz: (1) Solve with the Quadratic Formula (Example 1A). (2) Bonus: the same, with i in the answers (Example 1B). (3) How many roots, what type? (full Savvas quiz only) (4) Does a tossed ball reach a height? (full Savvas quiz only) (5) Which b give one real solution? (Example 5)

On the Topic 2 test: Question 12 asks you to solve $x^2 + 3x = -4$ by the formula; the answers have i (Example 1). Question 15 asks which b give $4x^2 + bx + 25 = 0$ one real solution (Example 5).

The Quadratic Formula. For any quadratic equation $ax^2 + bx + c = 0$ with $a \neq 0$:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The expression under the radical, $b^2 - 4ac$, is the **radicand**. It is also called the **discriminant**.

Example 1: Solve Quadratic Equations

What are the solutions to the equation? Write the equation as $ax^2 + bx + c = 0$, name a, b, c, then substitute. Use parentheses around negative numbers.

A. $3x^2 - 4x - 9 = 0$
$a = 3,\ b = -4,\ c = -9$

$$\begin{aligned} x &= \frac{-(-4) \pm \sqrt{(-4)^2 - 4(3)(-9)}}{2(3)} \\ &= \frac{4 \pm \sqrt{16 + 108}}{6} = \frac{4 \pm \sqrt{124}}{6} \\ &= \frac{4 \pm 2\sqrt{31}}{6} \qquad (\sqrt{124} = \sqrt{4 \cdot 31}) \\ &= \frac{2 \pm \sqrt{31}}{3} \end{aligned}$$

The radicand is $124 > 0$: **two real** solutions,

$$x = \frac{2 + \sqrt{31}}{3} \quad \text{and} \quad x = \frac{2 - \sqrt{31}}{3}.$$

B. $x^2 - 9x + 27 = 0$
$a = 1,\ b = -9,\ c = 27$

$$\begin{aligned} x &= \frac{-(-9) \pm \sqrt{(-9)^2 - 4(1)(27)}}{2(1)} \\ &= \frac{9 \pm \sqrt{81 - 108}}{2} = \frac{9 \pm \sqrt{-27}}{2} \\ &= \frac{9 \pm i\sqrt{27}}{2} \qquad (\sqrt{-1} = i) \\ &= \frac{9 \pm 3i\sqrt{3}}{2} \qquad (\sqrt{27} = 3\sqrt{3}) \end{aligned}$$

The radicand is $-27 < 0$: **two non-real** solutions,

$$x = \frac{9 + 3i\sqrt{3}}{2} \quad \text{and} \quad x = \frac{9 - 3i\sqrt{3}}{2}.$$

Say it: In Part B the radicand is _____. It is _____ (positive / negative), so the solutions are _____ (real / non-real) because ...

Try It 1: Solve using the Quadratic Formula

a. $2x^2 + 6x + 3 = 0$

$a =$ _____ $b =$ _____ $c =$ _____

$$x = \frac{-(\quad) \pm \sqrt{(\quad)^2 - 4(\quad)(\quad)}}{2(\quad)}$$

Radicand: _____

Real or non-real? _____

$x =$ _____

b. $3x^2 - 2x + 7 = 0$

$a =$ _____ $b =$ _____ $c =$ _____

$$x = \frac{-(\quad) \pm \sqrt{(\quad)^2 - 4(\quad)(\quad)}}{2(\quad)}$$

Radicand: _____

Real or non-real? _____

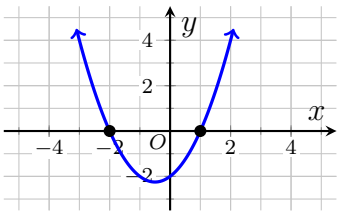
$x =$ _____

In part b, where does the i come from? Use the word *radicand*.

Challenge (DOK 3, self-paced). Seaira wants to use the Quadratic Formula to solve the equation $x^4 + 5x^2 - 5 = 0$. Is this possible? If so, describe the steps she should follow.

Key idea: Using the discriminant

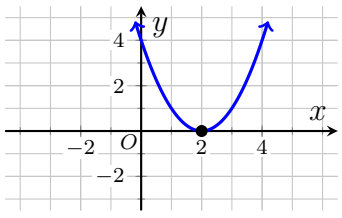
Predict the number and type of solutions using the **discriminant**, $b^2 - 4ac$ (the radicand of the Quadratic Formula).



$x^2 + x - 2 = 0$

$b^2 - 4ac > 0$

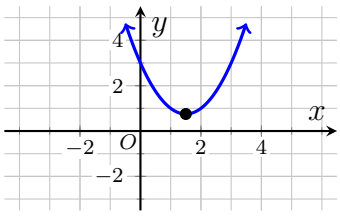
Two real solutions



$x^2 - 4x + 4 = 0$

$b^2 - 4ac = 0$

One real solution



$x^2 - 3x + 3 = 0$

$b^2 - 4ac < 0$

Two non-real solutions

Example 5: Use the Discriminant to Find a Particular Equation

What value(s) of b will cause $2x^2 + bx + 18 = 0$ to have one real solution?

For this equation, $a = 2$ and $c = 18$.
The equation will have a single rational solution when the discriminant is equal to 0.

$$\begin{aligned} b^2 - 4ac &= 0 \\ b^2 - 4(2)(18) &= 0 \\ b^2 - 144 &= 0 \\ b^2 &= 144 \\ b &= \pm 12 \end{aligned}$$

There are two possible equations:

$2x^2 + 12x + 18 = 0$ and $2x^2 - 12x + 18 = 0$.

Study tip. The equation $2x^2 + bx + 18 = 0$ will have two real solutions if $b > 12$ or $b < -12$. It will have two non-real solutions if $-12 < b < 12$.

Say it: The equation has one real solution when the discriminant is _____. There are _____ values of b that work because ...

Try It 5: Find the values of b

Determine the value(s) of b that ensure $5x^2 + bx + 5 = 0$ has two non-real solutions.

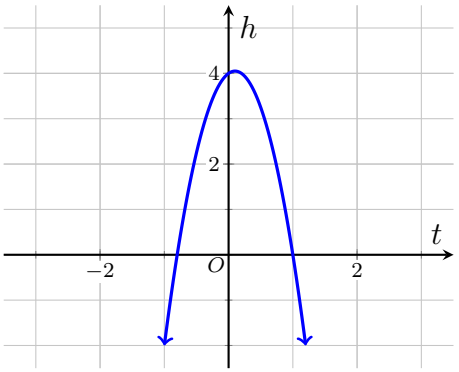
$a =$ ____ $c =$ ____
Two non-real solutions: the discriminant is _____ ($> 0, = 0, < 0$).
$b^2 - 4(\quad)(\quad)$ ____ 0

_____ $< b <$ _____

Say it: The solutions are _____ (real / non-real) when the discriminant is _____ ($> 0, = 0, < 0$), so b^2 must be less than _____ because ...

Practice: Will the ball reach that height?

Deon throws a ball into the air. The height, h, of the ball, in meters, at time t seconds is modeled by the function $h(t) = -5t^2 + t + 4$.



a. When will the ball hit the ground? (Hit the ground: $h = 0$.)

b. Will the ball reach a height of 5 meters?
Set $h(t) = 5$: _____
Standard form: _____ $= 0$
$a =$ ____ $b =$ ____ $c =$ ____ $b^2 - 4ac =$ _____
So:

IXL: Practice for the quiz

Do the starred skills first. **SmartScore 70 or above on each starred (★) skill replaces your quiz score.**

Code	Skill	Feeds
YQH ★	Solve a quadratic equation using the quadratic formula	Quiz 1; Quiz 2 (bonus). Example 1, Try It 1
QHK ★	Using the discriminant	Quiz 5. Example 5, Try It 5 (also Quiz 3, 4: full Savvas quiz only)
5VV	Introduction to complex numbers	Optional warm-up, no quiz score: square roots of negatives (Example 1B)
CEU	Solve quadratic equations: word problems	Optional: Practice page; Quiz 4 (full Savvas quiz only)

Prerequisites if you need review (no quiz score): CJC solve by factoring, NPH completing the square.

Before the quiz: Finish these sentences

To use the Quadratic Formula, first write the equation in the form _____ $= 0$.

The radicand of the Quadratic Formula, $b^2 - 4ac$, is called the _____.

If the discriminant is positive, the equation has _____. If it is zero, it has _____.

If the discriminant is negative, the equation has two _____ solutions, and the graph does not _____ the x-axis.

For one real solution, set the _____ equal to _____ and solve.